

Candidate Data Unification Pipeline — Design Specification

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1. Three-Layer Extensible Architecture

Layer 1 — Ingestion: Built on a dynamic adapter registry pattern. The abstract *SourceAdapter* exposes *detect()* and *extract()* -> *list[RawRecord]*. Adapters for Recruiter CSV, ATS JSON, PDF/Docx Resume (using *pypdf/docx*), notes, LinkedIn/GitHub URLs run isolated. A failure in an adapter returns an empty list, allowing other sources to merge without crashing the engine. New sources require zero changes to core engine logic.

Layer 2 — Engine: Orchestrates data through a sequential, decoupled stage pipeline: *Normalize* -> *Match* -> *Merge* -> *Confidence*. Identity matching executes in **linear O(N) time complexity** by building indexes of emails and normalized phones, eliminating nested pairwise comparison.

Layer 3 — Output/Projection: Completely decoupled presentation layer. Resolves nested data paths (e.g. *skills[].name*, *emails[0]*), renames fields, applies inline normalizers, and validates the output against configuration schemas. Prevents server crashes on invalid data by replacing failing types with *null*.

2. Canonical Schema & Field Normalization Rules

Ingested fields are normalized into uniform formats before resolving conflicts:

Field Name	Canonical Type	Normalization Rules / Formats
emails / phones	string[]	Emails: case-insensitive. Phones: E.164 (+[Country][Digits]) with US fallback (+1).
location / links	object	Location: {city, region, country: ISO-3166 alpha-2}. Links: {linkedin, github, portfolio, other}.
experience / education	object[]	Experience: start/end as YYYY-MM or 'Present'. Education: end_year as YYYY integer.
skills	object[]	Canonicalized name via synonym lookup table (e.g., js/javascript -> JavaScript). Confidence score attached.

3. Identity Matching & Merge Rules

- Identity Resolution:** Case-insensitive email is the primary key. If email is absent on either side, matching falls back to normalized phone. Names are never used as a sole key. If neither email nor phone is present, profiles stand alone.
- Priority-Based Conflict Resolution:** Fully configurable source precedence (default: *ATS JSON* > *Recruiter CSV* > *Resume* > *Notes*). A missing field in a higher-priority source never blocks lower-priority data from merging.
- Confidence Scoring:** Calculated as a weighted average across populated fields, weighted toward identity fields. Individual field confidence increases with corroboration (multiple agreeing sources) and source authority weights.
- Provenance:** Tracks origin metadata for every field (supplied source and extraction method), persisting independently of confidence.

4. Runtime Projection Config & Handled Edge Cases

Projection Config: Users pass a JSON configuration mapping output keys to canonical paths (e.g. *"primary_email": "emails[0]"*), setting missing policies (*null / omit / error*), and toggling confidence/provenance visibility.

- Duplicate Names:** Two candidates sharing a name with no email/phone will not match; they are kept separate.
- Empty Identifiers:** Sources with no email/phone are still ingested and stored as separate candidate profiles, rather than skipped.
- Inconsistent Spellings:** Skills like 'JS' and 'java script' normalize to 'JavaScript' before merging to prevent duplicate entries.
- Conflicting Emails:** Different emails with no shared phone will not auto-merge, creating separate profiles to prevent bad data blending.
- Out of Scope:** Fuzzy name matching, ML-based matching, and web scraping bypassing credentials are descope-declared for reliability.